Fan Degradation - Repair Manual

Symptoms: Increased vibration readings, reduced fan efficiency, abnormal noise during operation, elevated fan speed sensor deviation.

Common Causes:
- Foreign object damage (FOD) to fan blades
- Bearing wear in the fan shaft assembly
- Imbalance due to blade tip erosion

Recommended Actions:
1. Visually inspect fan blades for nicks, dents, or FOD damage
2. Check fan bearing temperature and vibration signature
3. Perform blade balance check - rebalance if deviation exceeds spec
4. Replace damaged blades immediately if crack propagation is detected
5. Verify fan speed sensor calibration matches actual RPM

Priority: HIGH - Fan imbalance can propagate to downstream stages and cause secondary damage if not corrected promptly.